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Unveiling the complexity of rifampicin drug susceptibility testing in *Mycobacterium tuberculosis*: comparative analysis with next-generation sequencing

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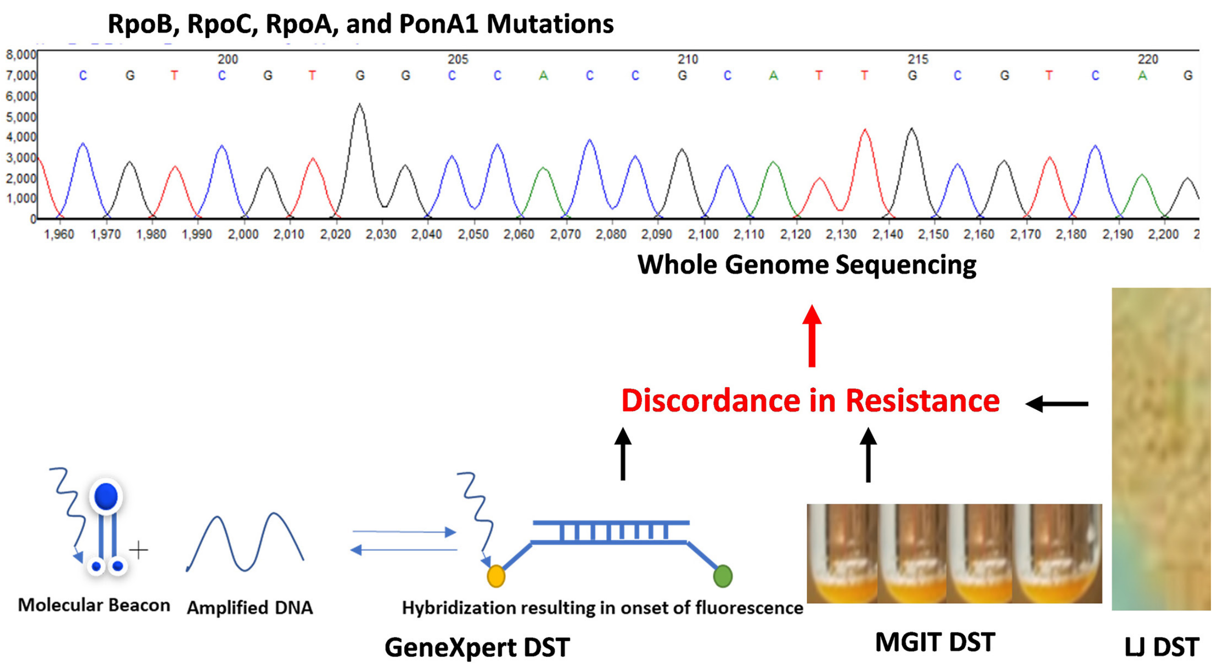

Graphical abstract

Abstract

Introduction. The discordance between phenotypic and molecular methods of rifampicin (RIF) drug susceptibility testing (DST) in *Mycobacterium tuberculosis* poses a significant challenge, potentially resulting in misdiagnosis and inappropriate treatment.

Hypothesis/gap statement. A comparison of RIF phenotypic and molecular methods for DST, including whole genome sequencing (WGS), may provide a better understanding of resistance mechanisms.

Aim. This study aims to compare RIF DST in *M. tuberculosis* using two phenotypic and molecular methods including the GeneXpert RIF Assay (GX) and WGS for better understanding.

Methodology. The study evaluated two phenotypic liquid medium methods [Lowenstein-Jensen (LJ) and Mycobacterium Growth Indicator Tube (MGIT)], one targeted molecular method (GX), and one WGS method. Moreover, mutational frequency in *ponA1* and *ponA2* was also screened in the current and previous RIF resistance *M. tuberculosis* genomic isolates to find their compensatory role.

Results. A total of 25 RIF-resistant isolates, including nine from treatment failures and relapse cases with both discordant and concordant DST results on LJ, MGIT and GX, were subjected to WGS. The phenotypic DST results indicated that 11 isolates (44%) were susceptible on LJ and MGIT but resistant on GX. These isolates exhibited multiple mutations in *rpoB*, including Thr444>Ala, Leu430>Pro, Leu430>Arg, Asp435>Gly, His445>Asn and Asn438>Lys. Conversely, four isolates that were susceptible on GX and MGIT but resistant on LJ were wild type for *rpoB* in WGS. However, these isolates possessed several novel mutations in the PonA1 gene, including a 10 nt insertion and two nonsynonymous mutations (Ala394>Ser, Pro631>Ser), as well as one nonsynonymous mutation (Pro780>Arg) in PonA2. The discordance rate of RIF DST is higher on MGIT than on LJ and GX when compared to WGS. These discordances in the Delhi/CAS lineages were primarily associated with failure and relapse cases.

Conclusion. The WGS of RIF resistance is relatively expensive, but it may be considered for isolates with discordant DST results on MGIT, LJ and GX to ensure accurate diagnosis and appropriate treatment options.

Impact Statement

The current study explains the complex landscape of rifampicin (RIF) resistance in *Mycobacterium tuberculosi*s through a detailed comparison of molecular and phenotypic drug susceptibility tests (DSTs) against whole-genome sequencing (WGS). The study highlighted the significant discordances by examining 25 WGS isolates with varying DST outcomes, particularly emphasizing the diagnostic challenges in detecting RIF resistance. Key findings include the identification of novel mutations associated with RIF resistance, underscoring the genetic diversity of resistant strains, and pinpointing compensatory mutations in non-*rpoB* genes (*ponA1* and *ponA2*) that may influence resistance phenotypes. The study contributes significantly to the field by providing a deeper understanding of the genetic mechanisms underpinning RIF resistance, which is crucial for refining diagnostic tests and treatment strategies. The breadth of interest spans clinical microbiology, infectious disease management and public health, offering critical insights for improving tuberculosis control programmes. This study represents a significant step beyond conventional DST to incorporate WGS for a more accurate assessment of drug resistance. The findings underscore the importance of integrating genomic data into tuberculosis diagnostics and of focusing on the evolutionary dynamics of drug resistance, enhancing the global efforts in tuberculosis control.

DATA AVAILABILITY

For current WGS sequence data: PRJNA953849 (SAMN34126561– SAMN34126585). For previous WGS sequence data: ERX-3360434–514, ERR-2510337–445, ERR-2510546–2645.

INTRODUCTION

Tuberculosis (TB) remains a major global health problem, affecting approximately 10 million people annually [1]. The emergence of drug-resistant *Mycobacterium tuberculosis*, the pathogen responsible for TB, presents a substantial challenge to global TB control efforts [2, 3]. Discordance in the diagnosis of drug resistance using drug susceptibility testing (DST) may have profound effects on TB control and transmission [4]. Specifically, the integration of genotypic DST for rifampicin (RIF) with whole-genome sequencing (WGS), as recently implemented in the UK, is strongly recommended for adoption in countries with a high burden of TB. This WGS solution offers rapid DST, furnishing clinically significant data to accurately

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Keywords: discordance; drug susceptibility testing; whole genome sequencing; *Mycobacterium tuberculosis*.
Abbreviations: DST, drug susceptibility testing; GX, GeneXpert RIF Assay; LJ, Lowenstein–Jensen; MDR, multidrug resistance; MGIT, Mycobacterium Growth Indicator Tube; RIF, rifampicin; RRDR, RIF resistance determining region; TB, tuberculosis; WGS, whole genome sequencing.
Four supplementary tables are available with the online version of this article.

detect resistance and compensatory gene mutations [5]. Among the primary drugs used in TB treatment, RIF and isoniazid (INH) constitute the cornerstone of standard therapy [6].

The discordant DST results using phenotypic and molecular methods, particularly in the case of RIF resistance, can potentially lead to mistreatment. These DST discordances, including INH resistance, RIF resistance, multidrug resistance (MDR) (resistance to INH and RIF), pre-XDR (extensively drug-resistant, i.e. resistant to RIF and INH and any fluoroquinolone), and XDR-TB (resistant to RIF, INH, any fluoroquinolone, and at least one of the drugs bedaquiline and linezolid) [3], are a major hurdle in TB control programmes. This is particularly serious in countries with high burdens of TB and significant cross-border traffic, including China and Pakistan [7].

Globally, 3.4% of new TB and 18% of previously treated cases are resistant to RIF or are MDR. In Pakistan, the estimated prevalence of new TB cases being MDR/RIF-resistant TB is 4.2% and among previously treated patients is 16% [8]. However, these estimates may be skewed if there is significant discordance in RIF DST.

Phenotypic DST using Lowenstein–Jensen (LJ) and liquid medium-based Mycobacteria Growth Indicator Tube (MGIT) is the WHO-recommended approach [9, 10]. However, technological development has led to target-specific molecular methods to detect specific mutations involved in resistance. Resistance to RIF is associated with mutations in *rpoB,* which encodes the β-subunit of the DNA-directed RNA polymerase (RNAP). The *rpoB* gene contains an 81 bp region (codons 426–452) called the RIF resistance determining region (RRDR) [9–12]. In targeted molecular DSTs such as GeneXpert (GX), only specific mutations within the RRDR are targeted to quickly identify resistance to RIF [10, 12–14].

Mutations outside the RRDR region may be less prevalent but cannot be ignored. Some other unknown mechanisms of RIF resistance, including compensatory mutations [15], necessitate phenotypic and WGS methods for better understanding. Some *rpoB* mutations (2–3%) that confer low-level RIF resistance are miscategorized as sensitive by phenotypic DST [16]. In such cases, mutations in *rpoA* or *rpoC* may compensate for mitigating the fitness cost in RIF-resistant TB [17, 18]. A previous study also reported evidence of 39 *M. tuberculosis* genomic areas, including *ponA1,* where mutations may confer directly or compensate for fitness costs associated with TB drug resistance [19].

The known mutations responsible for RIF resistance are directly involved in resistance. However, in 10–40% of *M. tuberculosis* clinically drug-resistant isolates, no known mutations have been identified. Additionally, even when known mutations are detected, other mutations in different genomic regions may still contribute to RIF resistance.

Our previous study investigated the discordance of RIF resistance based on LJ, MGIT and *rpoB* sequencing [20]. However, in continuation of this previous work, our goal here was to compare the results of RIF DST using two phenotypic methods (LJ and MGIT) and two genotypic methods (GX and WGS), to find mutations in other targets playing a compensatory role in RIF resistance. We selected 25 specific isolates exhibiting discordant and concordant DST results on LJ, MGIT and GX and processed them for WGS.

METHODS

Specimen processing

Sputum samples were decontaminated using previous methods [21, 22], followed by MGIT and LJ media cultures. About 0.1 ml of MGIT cultures were inoculated in TBc devices (Becton Dickinson) to confirm presence of the *M. tuberculosis* complex (MTBC). The MGIT DST was performed on MTBC as described previously [23]. The phenotypic RIF resistance was determined via MGIT using RIF at 1 $\mu g\,ml^{-1}$. The standard LJ DST was also performed [24] using RIF at 40 $\mu g\,ml^{-1}$. For quality assessment and accuracy, a susceptible H37Rv and two known resistant strains received from the Supra National Reference Laboratory (Antwerp, Belgium) [25] were used as quality controls. Finally, the *M. tuberculosis* isolates were subjected to GX DST (Cepheid GeneXpert System). The DST results were repeated twice before WGS analysis. Twenty-five RIF-resistant isolates discordant (phenotypically susceptible genotypically resistant or genotypically susceptible phenotypically resistant) and some concordance DST results (identical DST results on phenotypic and genotypic assays) as a control were selected for DNA isolation and WGS. Samples for WGS were selected based on specific objectives, including the need to investigate both concordant and discordant RIF resistance results across GX, MGIT and LJ methods. Discordant DST results among GX, MGIT and LJ were compared with WGS results to derive conclusive findings.

DNA isolation and WGS

M. tuberculosis DNA was isolated using the cetyltrimethylammonium bromide protocol [26]. Genomic DNA was subjected to WGS analysis at University College London using a 151 bp paired-end protocol in the Illumina MiSeq platform. The WGS results were collected in FASTQ format.

Further, to observe the evolutionary changes occurring in drug-resistant isolates, we analysed 289 previously sequenced drug-resistant whole genome sequences (ERX-3360434–514, ERR-2510337–445, ERR-2510546–2645) that were also retrieved from NCBI.

WGS analysis

The quality of retrieved WGS sequences was checked using FASTQC to trim low-quality raw reads. Sequences with GC bias and poor-quality sequence data (e.g. adapters in data that may easily interfere with sequence analyses) were removed.

All the WGS sequences were aligned with the reference genome (NC_000962.3) using the MTBSeq software package [27]. In brief, MTBSeq employs the BWA-mem algorithm for alignment, with subsequent processing of output data using the Samtools package to generate SAM/BAM formatted files. Variant calling was performed using the Genome Analysis Toolkit (GATK v.3.3.0) package [28].

RESULTS

Of 25 patients, 21 had a history of previous treatment, including nine with treatment failure (Table 1). Each patient's history indicated that discordance was more prevalent in cases with previous treatments for both pulmonary and extra-pulmonary TB (see File S1, available in the online version of this article). The concordant DST results served as controls to validate

Table 1. Sociodemographic data of the TB patients

S. no.	Lab sample ID (WGS)	Gender	Age (years)	History	TB category	Category	Outcome	TB (type)	Specimen
1.	MQ-01 (STAND 24)	M	25	Previously treated	CAT I	Relapse	Completed	P	S
2.	MQ-02 (STAND 25)	F	55	Previously treated	CAT II	Default	Lost to follow-up	P	S
3.	MQ-03 (STAND 26)	M	70	Never treated	NEW	New	New	P	S
4.	MQ-04 (STAND 27)	F	18	Previously treated	CAT II		Failure	EP	PF
5.	MQ-05 (STAND 28)	M	37	Never treated	NEW	New	New	P	S
6.	MQ-06 (STAND 29)	M	27	Previously treated	CAT II	Relapse	Treatment completed	P	S
7.	MQ-07 (STAND 30)	M	16	Previously treated	CAT II	Previously treated	Unknown	P	S
8.	MQ-08 (STAND 31)	F	25	Previously treated	CAT I	Relapse	Treatment completed	P	S
9.	MQ-09 (STAND 32)	M	19	Never treated	NEW	New	New	P	S
10.	MQ-10 (STAND 33)	F	30	Previously treated	CAT II	Failure	Failure	P	S
11.	MQ-11 (STAND 34)	M	14	Previously treated	CAT I	Failure	Failure	P	S
12.	MQ-12 (STAND 35)	F	48	Previously treated	CAT I	Failure	Failure	P	S
13.	MQ-13 (STAND 36)	F	23	Previously treated	CAT II	Failure	Failure	P	S
14.	MQ-14 (STAND 37)	M	24	Previously treated	CAT I	Default	Lost to follow-up	P	S
15.	MQ-15 (STAND 38)	F	46	Previously treated	CAT I	Relapse	Treatment completed	P	S
16.	MQ-16 (STAND 39)	M	49	Previously treated	CAT I	Failure	Failure	P	S
17.	MQ-17 (STAND 40)	F	29	Previously treated	CAT II	Relapse	Treatment completed	P	S
18.	MQ-18 (STAND 41)	M	25	Previously treated	CAT I	Relapse	Treatment completed	P	S
19.	MQ-19 (STAND 42)	F	40	Previously treated	CAT IV	Relapse	Treatment completed	P	S
20.	MQ-20 (STAND 43)	M	56	Previously treated	CAT I	Failure	Failure	P	S
21.	MQ-21 (STAND 44)	M	60	Previously treated	CAT I	Failure	Failure	P	S
22.	MQ-22 (STAND 45)	M	35	Unknown	CAT I	Relapse	Treatment completed	P	S
23.	MQ-23 (STAND 46)	M	21	Previously treated	unknown	Unknown	Unknown	P	S
24.	MQ-24 (STAND 47)	F	19	Previously treated	CAT I	Failure	Failure	P	S
25.	MQ-25 (STAND 48)	M	37	Previously treated	CAT I	Relapse	Treatment completed	P	S

Treatment type: P, pulmonary; EP, extrapulmonary. Sample: S, sputum; PF, pleural fluid.

the mutations associated with resistance across all four methods. Twenty-five *M. tuberculosis* isolates (Table 2), exhibiting discordant and concordant DST results on one targeted molecular method and two phenotypic methods (MGIT and LJ), were subjected to WGS to assess the DST accuracy. Nine samples demonstrated concordant GX, MGIT and LJ DST results, while 16 isolates exhibited discordant RIF resistance. Samples that showed resistance on any three DSTs (GX, MGIT, LJ) and exhibited discordant results, along with some isolates showing concordant results, were subjected to WGS. Results of WGS showed that all the discordant samples had RIF resistance *rpoB* mutations except four isolates (MQ-02, MQ-03, MQ-04 and MQ-06) (Table 2). The WGS data were submitted to NCBI GenBank under SRA: PRJNA953849 (Accession ID: SAMN34126561–SAMN34126585).

The frequency of each mutation in *rpoB* is provided in Table 3. The highest frequency mutations were His445>Asn (*n*=3) followed by Leu430>Pro (*n*=2), Asp435>Tyr (*n*=2) and Leu452>Pro (*n*=2).

Nine isolates (36%, 9/25) showed complete concordance in resistance with GX, LJ and MGIT (GX-R, LJ-R, MGIT-R) (Table 2). Four isolates (MQ-2, MQ-3, MQ-4 and MQ-6), susceptible on GX and resistant on LJ, were also identified as wild type (WT) on WGS. However, these WT harboured some synonymous mutations (Gly876>Gly and Ala1075>Ala) in the *rpoB* gene. These isolates were further investigated for compensatory mutations in *rpoA, rpoC* and other associated targets, including *ponA1* and *ponA2*.

Eleven isolates (44%) were susceptible on MGIT but had numerous mutations in the *rpoB* gene (Table 2). Six (6/11) isolates (MQ-5, 7, 8, 9, 13 and 21) showed susceptibility on both LJ and MGIT but resistance to RIF on GX and WGS. The specific mutations found were Thr444>Ala, Leu430>Pro, Leu430>Arg, Asp435>Gly, His445>Asn and Asn438>Lys, respectively. Sample MQ-8 (*rpoB*: Leu430>Arg), which was susceptible on LJ and MGIT DSTs but resistant on GX, also harboured one nonsynonymous mutation (Ala172>Val) and one synonymous mutation (Arg173>Arg) in *rpoC* (see File S2). Among the 25 WGS isolates, 20 were Delhi/CAS lineage, three Euro-American Superlineage, one Beijing lineage and one East African Indian lineage. Notably, when examining the discordance in MGIT DST, 10 out of the 11 discordant isolates were from the Delhi/CAS lineage. The remaining discordant isolate was from the East African Indian lineage. This suggests a higher prevalence of discordance in the Delhi/CAS lineage compared to other lineages (Table 2).

One synonymous mutation in *rpoA* (Asp324>Asp) and *rpoZ* (Leu46>Leu) was also detected in samples MQ-16 and MQ-19, respectively. Sample MQ-25, which is RIF-resistant, did not harbour any mutation in *rpoB*; however, it has one nonsynonymous (Val483>Gly) mutation in *rpoC* with concordant DST with all four DST methods (GX-R, LJ-R, MGIT-R and WGS-R). Three mutations (Phe424>Val, Pro358>Leu, Cys701>Trp) were detected outside the RRDR of *rpoB* (MQ-15 and MQ-23). Both isolates exhibited resistance on all four DST types.

The final DST results and their WGS sequencing are given in Table 2. Five of the final 25 RIF-resistant WGS isolates were susceptible on GX, six on LJ and 11 on MGIT.

Five isolates that were GX resistant and susceptible on LJ were also resistant on WGS. Isolate MQ-01 is susceptible on GX but resistant on LJ and MGIT, harbouring a large deletion (del 390: TGGACCAGAACAACCC).

Four isolates with RIF resistance (MQ-2, MQ-3, MQ-4 and MQ-6) did not harbour any mutation in *rpoB*. However, these isolates harboured numerous mutations in *ponA1* and *ponA2*. They harboured one nonsynonymous mutation in *ponA1* at Pro631>Ser, while in sample MQ-2, a 10 nt insertion GCCGCCGCCT at genome position 55540 was also detected in *ponA1*. The *ponA2* gene harboured one nonsynonymous mutation (Ala619Thr) in isolate MQ-2 (Table 4). Isolate MQ-3 harboured two nonsynonymous mutations (Ala394Ser, Pro631Ser) in *ponA1* and one nonsynonymous mutation (Pro780Arg) in *ponA2* (Table 4). One large insertion of 7 nt (GCCGCCT) was also detected in *ponA1* at genomic position 55543 in two isolates (MQ-3 and MQ-6). The WGS and mutation data of *ponA1* and *ponA2* are provided in File S3.

Six different nonsynonymous mutations (Thr444>Ala, Asp435>Gly, Leu430>Pro, His445>Asn, Leu430>Arg and Asn438>Lys) were detected in the six LJ susceptible isolates (MQ-5, MQ-7, MQ-8, MQ-9, MQ-13 and MQ-21) but they were GX resistant. Additionally, two synonymous mutations, Gly876>Gly and Ala1075>Ala, were found in combination with these nonsynonymous mutations. These two mutation combinations were the most common in 18 RIF-resistant WGS isolates.

Asn438>Lys (MQ-21) and Thr444>Ala (MQ-5) are novel discordant mutations, conferring resistance on GX but susceptibility on MGIT and LJ. Moreover, MQ-5 and MQ-21 harboured two nonsynonymous substitutions (Ala394Ser, Pro631Ser), one insertion in *ponA1* and one nonsynonymous substitution (Pro780Arg) in *ponA2*.

Five *M. tuberculosis* isolates, which were RIF-resistant on LJ, GX and MGIT, harboured double and triple combinations of *rpoB* mutations, including Met434>Val+His445>Asn, Leu430>Pro+Cys701>Trp, Pro358>Leu+Asp435>Tyr, His445>Tyr+Leu452>Pro and Phe424>Val+Ser441>Ala+His445>Asn. Two isolates (MQ-7 and MQ-22) (Table 2) were selected for comparison of Leu430>Pro. This mutation is always missed on MGIT and does not capture it as RIF resistance, but LJ DST sometimes captures it as resistance. In WGS, a combination of Leu430>Pro+Cys701>Trp (Table 2, MQ-22) was observed, in which Cys701>Trp has not been previously reported.

Table 2. Comparison of GeneXpert, LJ and MGIT results of RIF with whole genome sequencing of *rpoB*

S. no	Lab ID (WGS ID)	GX	LJ	MGIT	WGS_*rpoB* aa changed	*rpoB* mutation	Lineage	Previous study [20] R/S
1.	MQ-01 (STAND 24)	S	R	R	del 390 :TGGACCAGAACAACCC	Resistance	Delhi/CAS	Novel
2.	MQ-02 (STAND 25)	S	R	*NA	Ala1075Ala (**WT**)	No resistance	Beijing	NA
3.	MQ-03 (STAND 26)	S	R	R	Gly876Gly+Ala1075Ala (**WT**)	No resistance	Delhi/CAS	NA
4.	MQ-04 (STAND 27)	S	R	NA	WT	No resistance	Euro-American Superlineage	NA
5.	MQ-05 (STAND 28)	R	S	S	Thr444Ala+Gly876Gly +Ala1075Ala	Resistance	Delhi/CAS	Novel
6.	MQ-06 (STAND 29)	S	R	NA	Gly876Gly+Ala1075Ala (**WT**)	No resistance	Delhi/CAS	NA
7.	MQ-07 (STAND 30)	R	S	S	Leu430Pro+Gly876Gly+Ala1075Ala	Resistance	Delhi/CAS	6/14
8.	MQ-08 (STAND 31)	R	S	S	Leu430Arg+Ala1075Ala	Resistance	East African Indian	0/2
9.	MQ-09 (STAND 32)	R	S	S	Asp435Gly+Gly876Gly+Ala1075Ala	Resistance	Delhi/CAS	0/1
10.	MQ-10 (STAND 33)	R	R	S	Asp435Tyr+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	1/0
11.	MQ-11 (STAND 34)	R	R	S	Leu452Pro+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	15/5
12.	MQ-12 (STAND 35)	R	R	S	His445Leu+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	9/0
13.	MQ-13 (STAND 36)	R	S	S	His445Asn+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	7/11
14.	MQ-14 (STAND 37)	R	R	R	Met434Val+His445 Asn+Gly876 Gly + Ala1075Ala	Resistance	Delhi/CAS	5/0
15.	MQ-15 (STAND 38)	R	R	R	Phe424Val+Ser441 Ala+His445 Asn+Leu796Leu+Gly876Gly +Arg1061Arg+Ala1075Ala	Resistance	Delhi/CAS	Novel
16.	MQ-16 (STAND 39)	R	R	R	InDel of Phe at 434	Resistance	Euro-American Superlineage	Novel
17.	MQ-17 (STAND 40)	R	R	R	Ins of Arg at 432+Gly876Gly +Ala1075Ala	Resistance	Delhi/CAS	Novel
18.	MQ-18 (STAND 41)	R	R	S	(Agaa/A at 1308–10) Gln**436His+Asn437 del** Gly876Gly+Ala1075Ala	Resistance	Delhi/CAS	1/0
19.	MQ-19 (STAND 42)	R	R	R	Del of accagccagctg at 761 084	Resistance	Euro-American Superlineage	1/0
20.	MQ-20 (STAND 43)	R	R	S	Ser428Ile+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	1/0
21.	MQ-21 (STAND 44)	R	S	S	Asn438Lys+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	0/1
22.	MQ-22 (STAND 45)	R	R	R	Leu430Pro+Cys701 Trp+Gly876Gly+Ala1075Ala	Resistance	Delhi/CAS	6/14
23.	MQ-23 (STAND 46)	R	R	R	Pro358Leu+Asp435 Tyr+Gly876 Gly + Ala1075Ala	Resistance	Delhi/CAS	Novel
24.	MQ-24 (STAND 47)	R	R	R	His445Tyr+Leu452 Pro+Gly876 Gly + Ala1075Ala	Resistance	Delhi/CAS	8/0
25.	MQ-25 (STAND 48)	R	R	R	Ser450Leu+Gly876 Gly+Ala1075Ala	Resistance	Delhi/CAS	199/0

R, resistance; S, susceptible; NA, not applicable/contaminated; R/S, resistance/sensitive; Novel, first time from this geography.

Table 3. Frequency of each *rpoB* mutation in the current study

S. No.	Mutation	Frequency
1.	Phe424Val	1
2.	Ser428Ile	1
3.	Leu430Pro	2
4.	Leu430Arg	1
5.	Ins of Arg at 432	1
6.	Met434Val	1
7.	InDel of Phe at 434	1
8.	Asp435Gly	1
9.	Asp435Tyr	2
10.	Asn438Lys	1
11.	Ser441Ala	1
12.	His445Leu	1
13.	His445Asn	3
14.	His445Tyr	1
15.	Del of accagccagctg at 761084	1
16.	Leu452Pro	2
17.	Pro358Leu	1
18.	Cys701Trp	1

The Leu430>Arg sample (Table 2, MQ-08) was subjected to WGS because MGIT and LJ DSTs always declare it RIF sensitive. The MQ-08 harboured a large deletion (GCCGCCGCCT) in *ponA1* at genome position 55 540 (Table 4). Samples MQ-10 and MQ-23 were chosen for WGS to compare Asp435>Tyr as this mutant is always missed on MGIT but captured by LJ DST. However, these two isolates also harboured one large insertion of GCCGCCT at position 55 543 and two nonsynonymous mutations (Ala394>Ser and Pro780>Arg) in *ponA1* (Table 4). One isolate (Table 2, MQ-23), which was RIF-resistant on GX, LJ and MGIT, harboured a novel combination of mutation (Pro358>Leu+Asp435>Tyr). The MQ-11 isolate harboured Leu452>Pro, and was RIF susceptible on MGIT but resistant on LJ. However, combined with His445>Tyr in MQ-24 (Table 2), it was resistant on GX, MGIT and LJ DSTs. Similarly, sample MQ-12 harbouring His445>Leu was selected for WGS because it was susceptible on MGIT but RIF-resistant on LJ and GX. This sample harboured one large insertion and one nonsynonymous mutation (Table 4).

The His445>Asn mutation significantly conferred RIF resistance, particularly when present alongside other mutations. Isolate MQ-13 was susceptible on MGIT but harboured His445>Asn in *rpoB,* while in isolates MQ-14 and MQ-15, the His445>Asn in a combination with triple mutations (His445>Asn+Phe424>Val+Ser441>Ala) was resistant on LJ, GX and MGIT DST due to the presence of Phe424>Val+Ser441>Ala (Table 2). Similarly, in sample MQ-15, the His445>Asn+Met434>Val also showed RIF resistance on GX, LJ and MGIT DSTs due to Met434>Val+His445>Asn.

Mutation frequency in *ponA1* in previous drug-resistant genomic isolates

Based on the evidence of positive selection detected in 39 genomic locations within resistant isolates [22], we hypothesized that mutation frequencies in drug-resistant *M. tuberculosis* are increasing in some other targets, playing a compensatory role in resistance (Tables 4 and 5). For this, mutation frequency in *ponA1* among previous and current drug-resistant isolates was analysed. A total of 289 drug-resistant WGS sequences were screened in which only seven mutations were found in *ponA1*, of which six were detected in the N-terminus, including one synonymous mutation GCG/GCA (A167A) in the transglycosylase domain (Table 5). Most of these isolates were RIF-resistant [[[29]]]. One mutation [GAG/GCG (E519A)] was detected in the transpeptidase domain close to the C-terminal region of *ponA1*. Mutation CGT/CTT (R228L) was detected in two isolates between two domains. One substitution, ACG/ATG (T11M), close to the signal peptide was detected in two isolates.

Mutation frequency in RIF-resistant *ponA1* in current genomic isolates

Compared to the previous 289 drug-resistant *M. tuberculosis* WGS sequences, the genomic isolates of the current study showed high frequencies of *ponA1* mutations (Table 4). All current RIF-resistant isolates harboured the Pro631Ser (*n*=24) mutation

Table 4. Mutation frequency in the *ponA1* gene among the current WGS isolates exhibiting RIF resistance

S. no.	Lab ID (WGS ID)	GX	LJ	MGIT	WGS RpoB mutation	Compensatory mutation in PonA1
26.	MQ-01 (STAND 24)	S	R	R	del 390: TGGACCAGAACAACCC	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
27.	MQ-02 (STAND 25)	S	R	NA	Ala1075Ala (**WT**)	GCCGCCGCCT at 55540, Ala619Thr, Pro631Ser
28.	MQ-03 (STAND 26)	S	R	R	Gly876Gly+Ala1075Ala (**WT**)	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
29.	MQ-04 (STAND 27)	S	R	NA	**WT**	Pro631Ser
30.	MQ-05 (STAND 28)	R	S	S	Thr444Ala+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
31.	MQ-06 (STAND 29)	S	R	NA	Gly876Gly+Ala1075Ala (**WT**)	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
32.	MQ-07 (STAND 30)	R	S	S	Leu430Pro+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
33.	MQ-08 (STAND 31)	R	S	S	Leu430Arg+Ala1075Ala	Insertion of GCCGCCGCCT at 55540, Pro631Ser
34.	MQ-09 (STAND 32)	R	S	S	Asp435Gly+Gly876 Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
35.	MQ-10 (STAND 33)	R	R	S	Asp435Tyr+Gly876 Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Pro631Ser, Pro780Arg
36.	MQ-11 (STAND 34)	R	R	S	Leu452Pro+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser
37.	MQ-12 (STAND 35)	R	R	S	His445Leu+Gly876 Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser
38.	MQ-13 (STAND 36)	R	S	S	His445Asn+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
39.	MQ-14 (STAND 37)	R	R	R	Met434Val+His445 Asn+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
40.	MQ-15 (STAND 38)	R	R	R	Phe424Val+Ser441 Ala+His445Asn+Leu796 Leu+Gly876Gly+Arg1061Arg+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
41.	MQ-16 (STAND 39)	R	R	R	InDel of Phe at 434	Insertion of CCGT at 55553
42.	MQ-17 (STAND 40)	R	R	R	Ins of Arg at 432+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
43.	MQ-18 (STAND 41)	R	R	S	(Agaa/A at 1308–10) Gln**436His+Asn437 del**) Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
44.	MQ-19 (STAND 42)	R	R	R	Del of accagccagctg at 761084	Pro631Ser
45.	MQ-20 (STAND 43)	R	R	S	Ser428Ile+Gly876 Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
46.	MQ-21 (STAND 44)	R	S	S	Asn438Lys+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
47.	MQ-22 (STAND 45)	R	R	R	Leu430Pro+Cys701Trp+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg

Continued

Table 4. Continued

S. no.	Lab ID (WGS ID)	GX	LJ	MGIT	WGS RpoB mutation	Compensatory mutation in PonA1
48.	MQ-23 (STAND 46)	R	R	R	Pro358Leu+Asp435Tyr+Gly876Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
49.	MQ-24 (STAND 47)	R	R	R	His445Tyr+Leu452 Pro+Gly876 Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser, Pro780Arg
50.	MQ-25 (STAND 48)	R	R	R	Ser450Leu+Gly876 Gly+Ala1075Ala	Insertion of GCCGCCT at position 55543, Ala394Ser, Pro631Ser

Table 5. *ponA1* mutations in 289 previous *M. tuberculosis* genomic isolates

Genome ID	Position in genome	Position in gene	Ref	Alt	Locus tag	Codon	Effect
ERR2510365	54345	682	G	T	Rv0050	CGT/CTT (Arg228Leu)	Nonsyn
ERR2510366	53685	22	T	C	Rv0050	GTC/GCC (Val8Ala)	Nonsyn
ERR2510368	53694	31	C	T	Rv0050	ACG/ATG (Thr11Met)	Nonsyn
ERR2510374	55218	1555	A	C	Rv0050	GAG/GCG (Glu519Ala)	Nonsyn
ERR2510379	54345	682	G	T	Rv0050	CGT/CTT (Arg228Leu)	Nonsyn
ERR2510565	53694	31	C	T	Rv0050	ACG/ATG (Thr11Met)	Nonsyn
ERR2510610	53741	78	A	G	Rv0050	ACC/GCC (Thr27Ala)	Nonsyn

Nonsyn, Nonsynonymous.

except MQ-16. A novel insertion GCCGCCT at genomic position 55543 was detected in 20 RIF-resistant isolates. Similarly, mutations Ala394Ser (*n*=20) and Pro780Arg (*n*=17) were also detected at significant frequencies when compared with previous drug-resistant genomic isolates (File S3). None of the current *ponA1* mutations was found in previous genomes, which showed evolutionary changes in drug-resistant isolates, especially in RIF-resistant isolates. Mutations in other drug targets are detailed in File S4.

DISCUSSION

For identification of RIF resistance, both phenotypic and genotypic methods are recommended [30]. However, using different diagnostic platforms often results in discordant DST results, posing a dilemma for clinicians. In the current study, 25 isolates with discordant RIF DST results were subjected to WGS to investigate mutations in *rpoB,* conferring different results on different diagnostic platforms. Eleven isolates susceptible on MGIT (44%) harboured numerous mutations in *rpoB* (Table 2). One possible reason could be the presence of compensatory mutations and their role in RIF resistance. Some *rpoB* mutations might not significantly impact rifampicin binding, leading to a susceptible phenotype in DST despite the genetic alterations.

In 90–95% of RIF-resistant isolates, mutations are detected within the *rpoB* gene, primarily in the 81 bp region (codons 426–452) [9–11]. Although mutations in other regions of *rpoB* play a minor role, they cannot be ignored, as an outbreak with RIF resistance mutations outside the 81 bp region may still lead to misdiagnosis, mistreatment and increased transmission.

In our study, an isolate that harboured mutation Leu430>Arg was RIF-resistant on GX but susceptible on MGIT and LJ, while Leu430>Pro in combination with Cys701>Trp (Leu430>Pro+Cys701>Trp) was resistant (Table 4). A previous study [31] reported Leu430>Arg as RIF-resistant, while another study [32] reported it as a low-level RIF resistance mutation. Such isolates may have some compensatory mutations or Cys701>Trp combination, reducing RIF affinity, or a loss of fitness may cause slow growth in *M. tuberculosis* strains [33, 34], while MGIT is a fast-growth system. Cys701>Trp is outside the RRDR of *rpoB* and has been kept in the WHO mutation catalogue [35] as of 'Uncertain significance'. This discordance between LJ and MGIT may be due to the presence of Cys701>Trp and other compensatory mutations but not Leu430>Pro.

Mutation His445>Tyr in combination with Leu452>Pro demonstrated resistance on GX, LJ and MGIT, showing concordance with WGS; therefore, genotypic DST may also be performed for better understanding if the isolates are susceptible on MGIT [16]. Combinations of discordant mutations (Gln436>Hi+Asn437>del) (MQ-18, Table 2) and Ser428>Ile (MQ-20, Table 2) were MGIT susceptible, but GX and LJ resistant. Similar findings have been reported in previous studies [20]. However, these two isolates also harboured four mutations in *ponA1*, including a novel deletion (GCCGCCT) at position 55 543 (Table 4). Such mutations may play a compensatory role in RIF resistance, presenting challenges for DST and potentially leading to mistreatment.

The discordance was most commonly present in Delhi/CAS compared to other lineages. Table 2 shows that 10/11 MGIT discordances were in the Delhi/CAS lineage, and the remaining one was East African Indian (Table 2). However, as most of the current study isolates were Delhi/CAS (20/25), the association between discordance and lineage-specificity could not be established, nor has it been associated in previous studies.

In our study, four isolates (MQ-02, MQ-03, MQ-04 and MQ-06) were RIF-resistant on LJ but sensitive on GX. However, they did not harbour any mutation in *rpoB* and remained WT on WGS. These isolates harbour numerous mutations in *ponA1* and *ponA2,* including large nucleotide insertions. A previous study [19] demonstrated that *M. tuberculosis* is evolving mechanisms of drug resistance, acquiring genetic alteration in other targets. This prior study identified positive selection in 47 *M. tuberculosis* resistant genomes. By identifying convergent evolution, where mutations independently occur at the same nucleotide site or gene, the study successfully recovered 100% of established resistance markers, including mutation in *ponA1* [19].

Additionally, evidence of positive selection had been detected in 39 genomic locations within resistant isolates, encoding pathways related to bacterial cell wall biosynthesis, DNA repair and transcriptional regulation. These mutations may be involved directly in conferring drug resistance or compensating for the fitness costs associated with resistance. In one such case, functional genetic analysis of mutations in the *ponA1* gene showed an *in vitro* growth advantage in the presence of RIF [19]. However, these isolates require further investigation to gain a deeper understanding of RIF resistance.

To further investigate the role of *ponA1* in resistance, Farhat *et al*. conducted functional analyses on the mutant *ponA1* in the presence of RIF, demonstrating an *in vitro* growth advantage. The study observed that while known mutations contribute directly to resistance, 10–40% of clinically drug-resistant *M. tuberculosis* isolates do not have identifiable known mutations [36] as described here and, even if the known mutations were detected, there may be other mutations in other genomic regions contributing to drug resistance. Although the mutation type in *ponA1* differs from that in Farhat *et al*., its frequency in the current study is still significantly high (Table 4).

Asn438>Lys (Table 2, MQ-21) and Thr444>Ala (Table 2, MQ-5) are novel discordant mutations that conferred resistance on GX but susceptibility on MGIT and LJ. Both of these isolates harboured two nonsynonymous substitutions (Ala394Ser, Pro631Ser), one insertion in *ponA1* and one nonsynonymous substitution (Pro780>Arg) in *ponA2* (Table 4).

The clinical outcomes may differ for discordant DST results, and the patients have greater chances of mistreatment and increased transmission in the population. To explore additional information behind RIF resistance, WGS may be initiated for isolates exhibiting discordant DST results to reduce the chances of false negative resistance [37]. Recently, the UK has implemented WGS in TB diagnostics and observed its fast and substantial role in clinical settings compared to phenotypic DSTs. Moreover, geographically specific *rpoB* mutations may evade the current MGIT DST. Identifying the complete range of RIF resistance and compensatory mutations using WGS is more helpful for better management of TB burden [38]. In a programmatic setting, we would like to support the implementation of long-read WGS for MGIT and targeted genotypically susceptible isolates to better understand the discordance mutations associated with the compensatory role in RIF resistance. Long reads also detect structural variants, such as insertions, deletions, inversions and translocations, which are often challenging to identify using short-read sequencing technologies. Moreover, long reads improve mapping accuracy by reducing ambiguity in read mapping to the reference genome, and enhancing variant calling and annotation. It can also identify mixed infections and their respective resistance profiles.

Limitations of the current study

One limitation of this study is the relatively small sample size of WGS sequences. This sample size may not be sufficient to capture the full spectrum of *ponA1* mutation frequencies and their compensatory roles in resistance across diverse *M. tuberculosis* populations. Additionally, the study focuses primarily on *ponA1* mutations without considering potential interactions with other genetic factors that could contribute to drug resistance. Furthermore, the lack of functional assays to directly assess the impact of these mutations on RIF resistance limits the ability to draw definitive conclusions about their roles. Compared to previous changes, the evolutionary changes observed in the current isolates underscore the need for continuous and comprehensive genomic surveillance to fully understand the dynamics of drug resistance in *M. tuberculosis*.

CONCLUSION

Discordance in RIF resistance DST results was more common among patients with previous treatment, indicating potential challenges in accurately assessing drug resistance in such cases. WGS proved to be a valuable approach in validating mutations associated with resistance, particularly in cases where DST results were discordant across multiple methods. The study identified several mutations in the *rpoB* gene associated with RIF resistance, with some isolates harbouring novel mutations. Additionally, mutations in genes such as *ponA1* and *ponA2* were detected in isolates with discordant DST results, suggesting their potential role in compensatory mechanisms related to drug resistance. The study also highlighted lineage-specific differences in discordance rates, with the Delhi/CAS lineage showing a higher prevalence of discordance. These findings underscore the complexity of TB drug resistance and the importance of comprehensive molecular analysis for accurate diagnosis and treatment.

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Conflicts of interest
All authors have no conflicts of interest.

Ethical statement

This study was ethically approved by the Ethical Board of Quaid-i-Azam University Islamabad (QAU-BRC-2016). Samples and information were collected from TB patients at the National Tuberculosis Reference Laboratory, Islamabad, Pakistan.

Consent to publish

All individuals have given consent to participate in the study, or a parent or legal guardian.

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